



ESSAY

The perils of orthodontics

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I have suffered all of my life from the effects of poor orthodontics. I have had long term neck and shoulder pain as a result. The extraction of teeth has made me have to use a CPAP nightly. John Beck, MD. Far too often orthodontic treatment causes medical conditions when it is not done precisely. When done properly it can improve immune function, brain function, and optimize physical ability. Precision is necessary in orthodontics as the jaw nerve (trigeminal nerve) has 100 times more dense pain fibers than any other nerve in the body. When the jaw is not aligned precisely, this nerve becomes hyperactive leading to an elevation in the pain neurotransmitter (substance P). Substance P is known to cause a wide spectrum of conditions (neurogenic inflammatory disorders) and dysregulation of multiple systems (blood disorders, bone metabolism, vestibular dysfunction, hormone imbalance, etc.). Trigeminal nerve pain fibers mix with the vagus nerve and upper cervical nerves, thus potentially causing neck and internal organ pain. Research has shown that bite alignment therapy is 85% effective at eliminating headaches, regardless of the type of headache diagnosis. To perform orthodontics and jaw orthopedics (jaw alignment) with precision requires that the practitioner have knowledge of the correct theories of jaw function and an understanding of a process that is designed to achieve that end. Proper orthopedic theories, and hence an effective process, are not taught in most orthodontic programs in the United States. Orthodontists for the most part are taught how to get teeth straight but not how to assess jaw alignment, nor how to correct jaw misalignment.

The difference in our orthodontic treatment: To do things precisely, you cannot just put on braces and start moving teeth as is done by traditional orthodontists. To achieve a precise end, you need to align the jaw precisely before you start moving teeth. This tells you where and how much you need to move teeth. This is done with a diagnostic splint that is worn one to three months. The goal of orthodontists as they are taught in this country, is not to align the jaw but to create pretty, straight teeth. You cannot extract teeth in most cases, but rather you have to use functional appliances to grow the jaws bigger. You cannot surgically alter jaws in most cases as this cannot be done with the requisite precision. With the proper appliances surgery is rarely necessary.

It is seldom that you can treat with braces alone. Braces are not effective at growing back teeth longer. Deficient height in the back teeth is a problem found in the vast majority of orthodontic cases. Lack of height in the back teeth increases muscle tone throughout the body.

The process: Treatment time is typically slightly longer than traditional therapy, though in some cases much shorter.

Phase I: It is necessary in most cases, whether that be for orthodontics or pain management, to align the jaw precisely before moving teeth. This is done with an orthotic (splint) between the teeth. Typically, it needs to be adjusted a number of times before a stable position is achieved (1-6 months wear). Some patients need specially designed appliances to give them enough guidance to achieve stability in the proper orthopedic jaw relationship (e.g. twin blocks, herbst, omni, bionator, etc.).



Phase II: Once the jaw is aligned, definite treatment can be undertaken to develop jaw width, develop length, coordinate the teeth, and erupt them as necessary to establish a precise, solid bite. This treatment is often done with removable appliances, braces, or crowning of teeth depending on what is best and least invasive for the patient.

Phase III: Retentive phase - typically a night appliance for a varied amount of time.

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